Table R.1: Performance improvements of CoViP over the baseline across multi-image benchmarks. References are introduced in the author response.

Benchmark	Qwen3-VL-8B	CoViP	Δ (CoViP − Qwen3)
MM-NIAH [1]	86.7%	88.0%	+1.3%
MMNeedle [2]	47.3%	51.3%	+4.0%
MuirBench [3]	79.8%	81.8%	+2.0%
MMIU [4]	48.3%	49.8%	+1.5%
Average	**65.5%**	**67.7%**	**+2.2%**

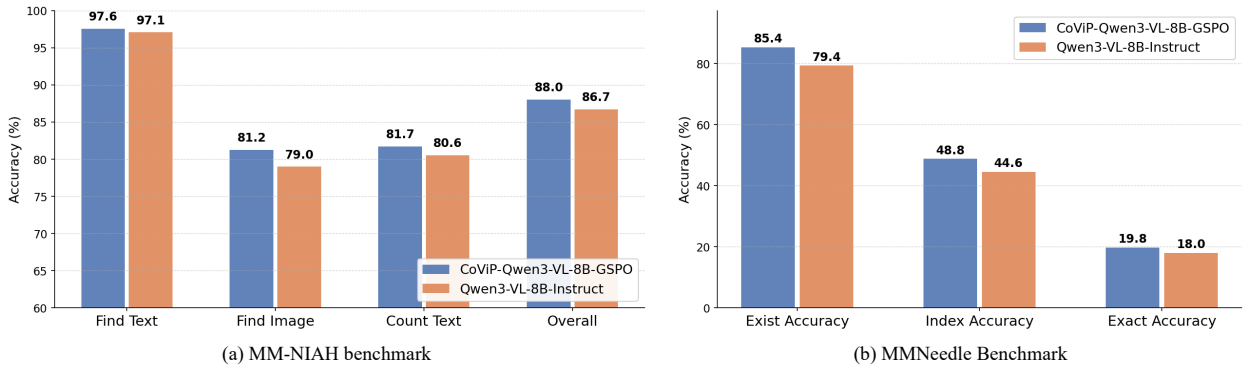
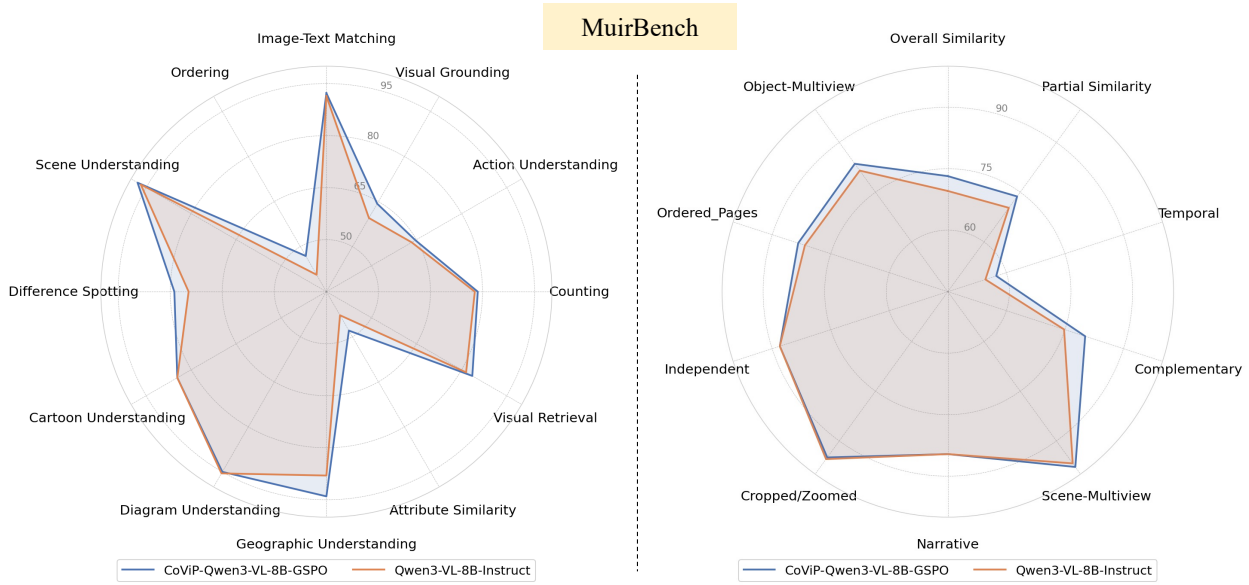
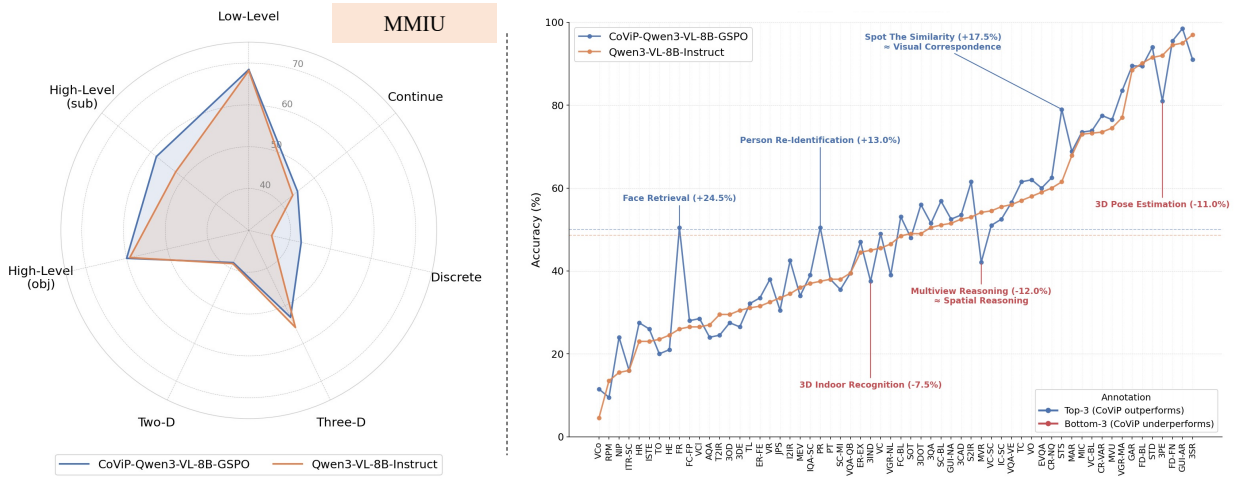


Figure R.1: Results on MM-NIAH (∼2.8K samples) and MMNeedle (∼1K subset, 4x4 grid), which evaluate long-context multimodal retrieval and needle localization in multi-image documents. CoViP consistently outperforms Qwen3-VL across all subtasks, with overall gains of +1.3% and +4.0% on MM-NIAH and MMNeedle, respectively.



(a) Task Accuracy (%) (b) Image Relation Accuracy (%)

Figure R.2: Results on MuirBench (∼1.3K answerable MCQs), which evaluates robust multi-image understanding across 12 tasks and 10 inter-image relation categories using a pairwise answerable/unanswerable design. Especially, CoViP shows gains on visual perception tasks (e.g., Visual Grounding, Image-Text Matching, Partial Similarity) than Qwen3-VL.



(a) Category Accuracy (%) (b) Per Task Accuracy (%)

Figure R.3: Results on MMIU (11K MCQs across 52 tasks), which evaluates multi-image relational understanding spanning temporal, semantic, and spatial categories. CoViP achieves consistent category-level gains and the largest per-task improvements on identity- and appearance-based retrieval tasks directly aligned with its RLVR objective (Face Retrieval: +24.5%, Visual Correspondence: +17.5%, Person Re-ID: +13.0%), while underperforming on 3D spatial reasoning tasks outside its training scope.